



For the analytic MHD equilibrium in the attachment, is the magnetic axis planar? Is there any vacuum rotational transform? (By that I mean if you took one of the flux surfaces, and computed the vacuum field inside that boundary shape, the rotational transform on the boundary would be nonzero).

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Yes—the MHD magnetic axis is planar. A separate vacuum calculation also gives a small, nonzero boundary rotational transform. For the outermost surface in the attachment, my numerical result is

$$\iota_{\text{vac}}(\partial\Omega_{1/200}) \simeq 0.00279554.$$

This is much smaller than the MHD transform on that same boundary, approximately 1.40908. The latter value is reported in the attachment; the vacuum value comes from a new calculation described below.

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1. The MHD magnetic axis is exactly planar

Equation (22) of the attachment gives

$$\gamma(t) = (a(S) \cos t, b(S) \sin t, 0).$$

Thus the axis is an ellipse in the plane $z = 0$, with semiaxes

$$a(S) \simeq 0.78193212, \quad b(S) \simeq 0.90078734.$$

Its torsion is identically zero. Its noncircular shape establishes non-axisymmetry, but not nonplanarity.

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2. The vacuum field inside the prescribed boundary has nonzero transform

I used your definition: select an original MHD surface $\psi = \psi_b$, keep its geometry fixed, and solve

$$\nabla \times \mathbf{B}_{\text{vac}} = 0, \quad \nabla \cdot \mathbf{B}_{\text{vac}} = 0 \quad \text{in } \Omega_{\psi_b}, \quad \mathbf{B}_{\text{vac}} \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega_{\psi_b}.$$

For a solid torus, this determines a one-dimensional space of vacuum fields, so fixing a nonzero toroidal flux or circulation determines the field uniquely. Its overall normalization does not affect the rotational transform. arXiv

For the attachment's parameters, I obtained the following. **Each row is a separate vacuum boundary-value problem**, and ψ_b denotes the original MHD label used to select the boundary geometry.

Selected boundary	Original MHD transform	Computed vacuum boundary transform
$\psi_b = 1/400$	$\sqrt{2} \simeq 1.41421356$	0.00274953
$\psi_b = 1/200$	1.40907746	0.00279554

The MHD column is from the attachment. Both vacuum values use its convention of increasing poloidal angle upward from the outboard side and increasing toroidal angle around the ellipse.

For the numerical calculation, I represented the field as

$$\mathbf{B}_{\text{vac}} = \nabla\phi + \nabla u, \quad \phi = \arg(x + iy),$$

with u approximated by a sum of point-source potentials whose sources all lie **outside** the solid torus. Each basis field therefore satisfies the two differential vacuum equations analytically. I fitted the coefficients to enforce boundary tangency, then calculated the boundary field-line return map.

For the outer boundary, the maximum independently sampled residual was

$$\frac{|\mathbf{B}_{\text{vac}} \cdot \mathbf{n}|}{|\mathbf{B}_{\text{vac}}|} < 5 \times 10^{-10}.$$

Refining the source discretization changed the computed transform from 0.0027955378 to 0.0027955391. These are numerical convergence checks, not certified error bounds, but they give clear evidence that the transform is nonzero.

3. Why a planar axis does not eliminate the vacuum transform

There is also an analytic explanation in the thin-torus limit. The relevant geometry is not just the axis: **the cross-sections tilt and change elongation as one moves toroidally**. The near-axis expression for vacuum transform includes this cross-sectional contribution even when axis torsion vanishes. [arXiv](#)

For this particular family, that contribution can be calculated explicitly.

Write

$$h = \sqrt{4S^2 + \varepsilon^2}, \quad G(t) = \frac{h + \varepsilon \cos 2t}{2}, \quad V(t) = \frac{\varepsilon}{2} \sin 2t,$$

and let $k = \sqrt{2\psi_b}$. Expanding the attachment's surface embedding about its magnetic axis gives, in the plane normal to the axis,

$$\begin{pmatrix} X \\ z \end{pmatrix} = k \begin{pmatrix} A(t) & C(t) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \chi \\ \sin \chi \end{pmatrix} + O(k^2),$$

where X is displacement along the outward horizontal normal, and

$$A(t) = \frac{\sqrt{G(t)}}{h} \operatorname{sech} V(t), \quad C(t) = \frac{\sqrt{G(t)}}{h} \tanh V(t).$$

This follows by expanding equations (16), (17), and (21) of the attachment. In particular, the nonzero $C(t)$ is the cross-sectional tilt. [20260911-01_Astra_analytic_3d_e... +1](#)

At leading order in the tube thickness, the curl-free condition makes the transverse field-line matrix symmetric in this nonrotating normal frame. Requiring it to transport the ellipses above gives

$$\frac{d\chi}{dt} = \frac{AC' - CA'}{1 + A^2 + C^2} = \frac{G(t)}{h^2 + G(t)} \operatorname{sech} V(t) V'(t),$$

where primes denote t -derivatives. Hence my thin-torus calculation gives

$$\lim_{\psi_b \rightarrow 0} \iota_{\text{vac}}(\partial\Omega_{\psi_b}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{G(t)}{h^2 + G(t)} \operatorname{sech} V(t) V'(t) dt.$$

This integral is **strictly positive for $\varepsilon > 0$** . To see the sign without relying on numerical quadrature, use

$$\frac{d}{dt} \arctan(\sinh V) = \operatorname{sech} V V'$$

and integrate by parts:

$$\iota_{\text{vac}}^{\text{thin}} = \frac{\varepsilon h^2}{2\pi} \int_0^{2\pi} \frac{\sin 2t \arctan \left[\frac{\sinh(\frac{\varepsilon}{2} \sin 2t)}{\frac{\varepsilon}{2}} \right]}{[h^2 + G(t)]^2} dt > 0.$$

The two factors in the numerator have the same sign and are nonzero except at isolated values of t .

For the parameters in the attachment,

$$\iota_{\text{vac}}^{\text{thin}} \simeq 0.0027040156,$$

consistent with the two finite-boundary calculations above.

The physical interpretation is that the cross-sections **rock back and forth rather than making a complete rotation**, but their elongation changes in a correlated way. The two directions of rocking therefore do not cancel in their contribution to field-line winding. No nonzero axis torsion is required.

Scope of these results

The exact planar ellipse is the axis of the **original MHD field**. I have not located the magnetic axis of the recomputed finite-radius vacuum field, nor established that it retains nested interior flux surfaces. Neither assumption was used in calculating its boundary transform.

Likewise, the thin-torus limit above concerns a sequence of shrinking-boundary vacuum problems; it is not the on-axis transform of the vacuum field inside the fixed outermost boundary.

The Python scripts, solved vacuum fields, and numerical results reproduce the boundary calculations and convergence checks.

Sources



This class of analytic 3d equilibria would be more interesting if they had larger vacuum rotational transform, which could come from either average torsion of the magnetic axis or elongation that rotates poloidally in a consistent direction as the torus is traversed toroidally. Can this family of analytic solutions be generalized to increase the vacuum rotational transform, while remaining in equilibrium?

Yes. I found an exact extension that does this through consistently rotating elongation, without making the MHD magnetic axis nonplanar. A concrete example has

$$\iota_{\text{vac}}(\text{boundary}) \simeq 0.176405,$$

from a separate finite-radius vacuum calculation. Stronger shaping gives thin-torus vacuum transforms of approximately 0.24, 0.43, and higher.

The extension introduces an adjustable vertical wavenumber λ . The important point is that **both the field components and the pressure must change with λ** ; simply replacing e^{iz} by $e^{i\lambda z}$ would not suffice. Below, the MHD equilibrium is exact. The vacuum transform is evaluated analytically at leading order in the tube thickness and numerically for a finite boundary.

1. An exact extension of the attached solution

Keep the attachment's definitions

$$\zeta = x + iy, \quad K = \bar{\zeta} \sqrt{1 + \frac{\varepsilon}{\bar{\zeta}^2}}, \quad \Phi = \zeta K + \frac{\pi}{2} - S, \quad w = \frac{i\kappa}{2K} \sin \Phi.$$

The square root is still taken around the expression inside the parentheses.

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For any constant $\lambda > 0$, define

$$\begin{aligned} B_x + iB_y &= e^{i\lambda z} w, \\ B_z &= -\frac{\kappa}{\lambda} \operatorname{Re}(e^{i\lambda z} \cos \Phi), \\ \psi &= \frac{1}{2} \left[\sin^2(\lambda z) + \left(\frac{\lambda B_z}{\kappa} \right)^2 \right], \\ p &= p_a - \frac{\kappa^2}{\lambda^2} \psi. \end{aligned} \tag{1}$$

The attached solution is the case $\lambda = 1$.

Equivalently, if $\mathbf{B}_1, \psi_1$ denote the original solution at the same ε, S, κ , this is

$$\mathbf{B}_\lambda(x, y, z) = \begin{pmatrix} B_{1x}(x, y, \lambda z) \\ B_{1y}(x, y, \lambda z) \\ \lambda^{-1} B_{1z}(x, y, \lambda z) \end{pmatrix}, \quad \psi_\lambda(x, y, z) = \psi_1(x, y, \lambda z). \tag{2}$$

Thus, geometrically, the surfaces are compressed vertically by a factor λ . **Such a deformation would not preserve a generic MHD equilibrium. It works here because of the special separability of the acceleration potential.**

Why force balance survives

The three identities for w used in the attachment remain unchanged:

$$w_{\zeta\zeta} = -K^2 w, \quad w_{\zeta\bar{\zeta}} = -|\zeta|^2 w, \quad ww_{\zeta\zeta} - w_\zeta^2 = \frac{\kappa^2}{4}.$$

These are the identities underlying the original force-balance proof.

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Repeating that proof with (1) gives

$$\nabla \cdot \mathbf{B}_\lambda = 0$$

and

$$(\mathbf{B}_\lambda \cdot \nabla) \mathbf{B}_\lambda = -\nabla Q_\lambda, \quad Q_\lambda = -\frac{|w|^2}{2} + \frac{\kappa^2}{2\lambda^2} \sin^2(\lambda z).$$

Since

$$Q_\lambda + \frac{|\mathbf{B}_\lambda|^2}{2} = \frac{\kappa^2}{\lambda^2} \psi,$$

it follows exactly that

$$(\nabla \times \mathbf{B}_\lambda) \times \mathbf{B}_\lambda = \nabla p, \quad \mathbf{B}_\lambda \cdot \nabla \psi = 0. \tag{3}$$

This is an exact Euclidean, scalar-pressure equilibrium—not an asymptotic equilibrium construction.

An additional useful property is that λ **changes the geometry and vacuum transform without changing the MHD rotational transform at fixed ψ** . The field-line systems before and after (2) are conjugate by the vertical compression. Equivalently, with

$$P = -\frac{\lambda B_z}{\kappa}, \quad Z = \sin(\lambda z),$$

the equations for (P, Z, t) are independent of λ , just as the original angular equations contain no such parameter.

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2. The larger-shaping examples remain regular nested-torus equilibria

There is a domain issue worth making explicit before increasing ε .

The attachment restricts to $x^2 + y^2 > \varepsilon$ to make the square-root branch unambiguous. That restriction is **sufficient but not necessary**. 20260911-01_Astra_analytic_3d_e... The same formula for K is analytic throughout the complement of the focal segment

$$\{x = 0, |y| \leq \sqrt{\varepsilon}\}. \quad (4)$$

Away from this segment, the expression inside the principal square root never crosses its negative-real-axis cut or vanishes.

Every positive- σ confocal ellipse used in the attachment lies outside that segment. Hence the construction extends to more elongated axis ellipses than the original sufficient domain restriction allowed.

Specifically, retain the attachment's function $\sigma(P, Z, t)$, but replace its position map by

$$\mathbf{r}_\lambda(P, Z, t) = \left(a(\sigma) \cos t, b(\sigma) \sin t, \frac{\arcsin Z}{\lambda} \right). \quad (5)$$

Here

$$h(\sigma) = \sqrt{4\sigma^2 + \varepsilon^2}, \quad a(\sigma) = \sqrt{\frac{h(\sigma) - \varepsilon}{2}}, \quad b(\sigma) = \sqrt{\frac{h(\sigma) + \varepsilon}{2}},$$

and $\sigma(P, Z, t)$ is precisely the inverse formula in the attachment. 20260911-01_Astra_analytic_3d_e... +1

A sufficient condition for the solid torus $P^2 + Z^2 \leq k_b^2$ is

$$0 < k_b < 1, \quad S > \arcsin k_b. \quad (6)$$

One can show that $|\sigma - S| \leq \arcsin k_b$, so σ stays positive. The Jacobian is the original positive Jacobian divided by λ :

$$\det \frac{\partial(x, y, z)}{\partial(P, Z, t)} = \frac{G(\sigma, t)}{\lambda h(\sigma) \sqrt{\cosh^2 V - P^2 - Z^2} \sqrt{1 - Z^2}} > 0. \quad (7)$$

Together with global injectivity, this establishes nested, embedded, analytic toroidal surfaces. Since

$$\psi = \frac{P^2 + Z^2}{2},$$

the pressure gradient vanishes only on the magnetic axis.

The MHD axis remains

$$\gamma(t) = (a(S) \cos t, b(S) \sin t, 0). \quad (8)$$

It is therefore still planar and, for $\varepsilon > 0$, noncircular. The attachment's argument excluding axisymmetry still applies. 20260911-01_Astra_analytic_3d_e... +1

3. Choosing λ to make the elongation rotate continuously

This is the main geometric result.

For this section, write

$$h = \sqrt{4S^2 + \varepsilon^2}, \quad G(t) = \frac{h + \varepsilon \cos 2t}{2}, \quad V(t) = \frac{\varepsilon}{2} \sin 2t.$$

Let X denote displacement along the outward horizontal normal to the elliptical axis. Expanding (5) on $\psi = k^2/2$ gives the normal-plane cross-section

$$\begin{pmatrix} X \\ z \end{pmatrix} = kL(t) \begin{pmatrix} \cos \chi \\ \sin \chi \end{pmatrix} + O(k^2),$$

where

$$L(t) = \begin{pmatrix} \frac{\sqrt{G}}{h} \operatorname{sech} V & \frac{\sqrt{G}}{h} \tanh V \\ 0 & \lambda^{-1} \end{pmatrix}. \quad (9)$$

Define

$$r(t) = \frac{\lambda \sqrt{G(t)}}{h}.$$

Apart from an overall scalar factor, the ellipse's shape matrix is

$$C(t) = \begin{pmatrix} r^2 & r \tanh V \\ r \tanh V & 1 \end{pmatrix}. \quad (10)$$

Its major-axis orientation $\beta(t)$ is determined by

$$2\beta(t) = \arg[(r^2 - 1) + 2ir \tanh V],$$

with the argument unwrapped continuously.

Now choose

$$\boxed{\lambda^2 = 2h.} \quad (11)$$

Then, with $q = \varepsilon/h \in (0, 1)$,

$$r^2 = 1 + q \cos 2t,$$

so the vector determining twice the orientation is

$$\left(q \cos 2t, 2\sqrt{1 + q \cos 2t} \tanh \left[\frac{\varepsilon}{2} \sin 2t \right] \right). \quad (12)$$

This vector winds twice around the origin as t goes from 0 to 2π . Consequently, the major axis rotates through 2π : one complete poloidal rotation per toroidal circuit.

Moreover, the rotation is monotonic, not merely a net winding:

$$\boxed{\beta'(t) > 0.} \quad (13)$$

For example, the continuously unwrapped major-axis orientations at

$$t = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$$

are respectively

$$\beta = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi.$$

The positivity proof for the derivative is included in the linked derivation.

Thus, the extension produces exactly the mechanism you suggested: **elongation rotating poloidally in a consistent direction**, despite zero torsion of the MHD axis.

4. The resulting vacuum transform

The rotating-cross-section contribution is the usual near-axis geometric mechanism for vacuum transform; it does not require nonzero axis torsion. The near-axis formulation in Jorge, Sengupta & Landreman includes this dependence on ellipse orientation and elongation. [✖ arXiv](#)

For the present family, I obtain the following explicit leading thin-tube formula:

$$\iota_{\text{vac}}^{\text{thin}} = \frac{1}{2\pi} \int_0^{2\pi} \frac{\lambda^2 G(t)}{h^2 + \lambda^2 G(t)} \operatorname{sech} V(t) V'(t) dt. \quad (14)$$

Here $\iota_{\text{vac}}^{\text{thin}}$ is the leading value for the sequence of separate vacuum problems inside boundaries $\psi = \psi_b$ as $\psi_b \rightarrow 0$. It is **not** an exact finite-radius formula.

To see where (14) comes from, write the leading transverse vacuum field-line system as

$$\mathbf{x}' = M(t)\mathbf{x}.$$

The normal frame of the planar axis has no rotation about the tangent. Curl-freeness therefore makes M symmetric. Transporting the ellipses (9) requires

$$M = L'L^{-1} + \omega LJL^{-1}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

For

$$L = \begin{pmatrix} a & c \\ 0 & d \end{pmatrix}, \quad d = \lambda^{-1},$$

the symmetry condition gives

$$\omega = \frac{ac' - ca'}{a^2 + c^2 + d^2}.$$

Substitution of (9), followed by averaging over t , gives (14).

For the monotonically rotating choice (11), there is also an illuminating equivalent expression:

$$\iota_{\text{vac}}^{\text{thin}} = \frac{1}{2\pi} \int_0^{2\pi} \beta'(t) \frac{[E(t) - 1]^2}{E(t)^2 + 1} dt, \quad (15)$$

where $E(t) \geq 1$ is the local ellipse elongation. This makes two points transparent: monotonic rotation gives positive transform, but **rotating an almost circular cross-section still gives very little transform**.

Numerical values of the thin-tube expression

These values are evaluations of (14):

ε	S	λ	$\iota_{\text{vac}}^{\text{thin}}$
0.2	0.70435455	1, original example	0.00270402
0.2	0.70435455	1.686912, rotating elongation	0.00352279
2	1	2.378414, rotating elongation	0.175574
2	0.5	2.114743, rotating elongation	0.237852
4	0.5	2.871622, rotating elongation	0.431077

Thus, adjusting λ alone at the original weak shaping produces only a modest increase. **The substantial increase comes from combining stronger shaping with the appropriate λ .**

There is even a useful analytical limit. Within the thin-tube calculation, keeping $S > 0$ fixed and choosing $\lambda^2 = 2h$ gives

$$\iota_{\text{vac}}^{\text{thin}} \longrightarrow \frac{2}{3} \quad \text{as } \varepsilon \rightarrow \infty. \quad (16)$$

That limit involves increasingly extreme geometry. In particular, at $t = \pi/4$,

$$E = e^{\varepsilon/2}.$$

It demonstrates that the enhancement is not confined to a small perturbative value, but it is not a recommendation to take the shaping arbitrarily far.

5. A finite-radius example, using your definition of vacuum transform

To check that the enhancement is not just a thin-tube artifact, I used

$$\varepsilon = 2, \quad S = 1, \quad \lambda = 2^{5/4} \simeq 2.37841423, \quad \psi_b = \frac{1}{200}. \quad (17)$$

These parameters satisfy the regular-domain condition (6), with $k_b = 0.1$.

For the boundary supplied by the **exact MHD equilibrium**, I separately solved

$$\nabla \times \mathbf{B}_{\text{vac}} = 0, \quad \nabla \cdot \mathbf{B}_{\text{vac}} = 0, \quad \mathbf{B}_{\text{vac}} \cdot \mathbf{n} = 0,$$

with nonzero toroidal circulation. The resulting boundary transform is

$$\iota_{\text{vac}}(\partial\Omega_{1/200}) \simeq 0.1764050176. \quad (18)$$

The thin-tube prediction is 0.1755737214.

The vacuum calculation used harmonic point-source potentials with all sources outside the torus. On an independent boundary-check grid, the finest solve had

$$\max \frac{|\mathbf{B}_{\text{vac}} \cdot \mathbf{n}|}{|\mathbf{B}_{\text{vac}}|} \simeq 2.5 \times 10^{-9}.$$

Two source-discretization refinements gave transforms 0.176405017618 and 0.176405017590; doubling the boundary interpolation grid gave 0.176405017594. These are convergence checks, not certified error bounds.

For comparison, the **MHD** boundary transform in this example is

$$\iota_{\text{MHD}}(\psi_b) \simeq 2.2877717355,$$

which is a different quantity and remains independent of λ .

I have not assumed that the finite-radius vacuum field retains the original MHD axis or the original interior flux surfaces. Only its prescribed boundary is used in this calculation.

Conclusion

There is an exact, explicit route to substantially larger vacuum transform within an extension of this family. It preserves the analytic nested surfaces, scalar-pressure force balance, nonzero pressure gradient off the axis, and non-axisymmetry. The MHD axis remains planar, but the choice

$$\lambda^2 = 2\sqrt{4S^2 + \varepsilon^2}$$

makes the near-axis elongation rotate monotonically through a full turn, and stronger shaping then produces appreciable vacuum transform.

The [full derivation](#) includes the domain and monotonic-rotation proofs. The [reproducible code and numerical results](#) include the exact Cartesian field, automatic-differentiation checks of equilibrium, vacuum solves, and convergence tests.

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